

Logistic Regression

Beginner-friendly notes: concept, intuition, sigmoid function, probability, classification, and the basic mathematical workflow.

1. What is Logistic Regression?

Logistic Regression is mainly used for **classification problems**. It predicts the probability that an observation belongs to a particular class.

Examples: Will a customer buy or not? Is an email spam or not spam? Will a student pass or fail? Is a customer likely to churn or stay?

Typical binary classes: 0 = No, 1 = Yes

2. Linear Regression vs Logistic Regression

Linear Regression	Logistic Regression
Used for regression	Used for classification
Predicts continuous values	Predicts class probabilities
Output can be any number	Probability is between 0 and 1
Example: house price	Example: spam / not spam
Uses a straight-line equation	Uses the Sigmoid function

3. What does Logistic Regression Predict?

Instead of directly predicting a class, Logistic Regression first predicts a **probability**.

Example: Probability of a customer buying = 0.82

We can then convert that probability into a class using a threshold:

Probability $\geq 0.5 \rightarrow$ Class 1

Probability $< 0.5 \rightarrow$ Class 0

4. Why not simply use Linear Regression?

For classification, we usually want probabilities between 0 and 1. Linear Regression can produce values such as -0.7, 1.2, or 2.5, which are not valid probabilities.

Therefore, Logistic Regression uses a special function called the **Sigmoid Function** to convert the model's output into a value between 0 and 1.

5. Sigmoid Function

Formula: $\sigma(z) = 1 / (1 + e^{-z})$

The Sigmoid function has an S-shaped curve. It takes any real-valued input and converts it into a number between 0 and 1.

z	Sigmoid / Probability
-5	0.0067
-2	0.119
0	0.500
2	0.881
5	0.993

6. Logistic Regression Equation

Step 1: Linear equation

$$z = b_0 + b_1X_1 + b_2X_2 + \dots + b_nX_n$$

Step 2: Apply Sigmoid

$$P(Y = 1) = 1 / (1 + e^{-z})$$

So the overall process is:

Input Features → **Linear Equation** → **z** → **Sigmoid** → **Probability** → **Threshold** → **Class 0/1**

7. Simple Example: Student Pass/Fail

Suppose we want to predict whether a student will pass based on the number of hours they study.

Study Hours	Probability of Passing	Prediction
1	0.08	0 = Fail
2	0.18	0 = Fail
3	0.35	0 = Fail
4	0.55	1 = Pass
5	0.73	1 = Pass
6	0.88	1 = Pass

Using the default threshold of 0.5, probabilities of 0.5 or higher are classified as 1, while probabilities below 0.5 are classified as 0.

8. Complete Workflow

1. Give the model input features.
2. Calculate the linear combination: $z = b_0 + b_1X_1 + \dots + b_nX_n$.
3. Pass z through the Sigmoid function.
4. Get a probability between 0 and 1.
5. Apply a classification threshold.
6. Produce the final class prediction: 0 or 1.

9. Important Terms to Remember

Term	Meaning
Classification	Predicting a category or class.
Probability	Likelihood of belonging to a class, between 0 and 1.
Sigmoid	Function that converts z into a value between 0 and 1.
Threshold	Cutoff used to convert probability into a class.
Class 0	Usually represents No / Negative class.
Class 1	Usually represents Yes / Positive class.

10. Quick Revision

Logistic Regression = Classification model

It first calculates a linear equation, then uses the Sigmoid function to obtain a probability, and finally uses a

threshold to predict a class.

Remember: Linear equation $\rightarrow$ Sigmoid $\rightarrow$ Probability $\rightarrow$ Threshold $\rightarrow$ Class